

Low-Cost Velodrome Lap Timing via iPhone and Computer Vision: Report

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Track cyclists rely on accurate lap timing during training sessions to gauge performance, effort, and improvement. However, current solutions such as GPS and race-grade timing infrastructure are impractical and/or unavailable. This project outlines a low-cost lap-timing system that only requires a smartphone (iPhone) and a phone mount. The phone is mounted on the handle bar face down, and the device's camera is used to detect start/finish line crossings. The system uses a computer vision pipeline running on-device with a retrained YOLOv26n model to identify the start/finish line, timestamp each crossing, compute lap splits, and provide real-time audio feedback to the rider. The app also enables users to export training data to Strava (a social media and fitness tracking app). Given the nature of track cycling, the system must perform well at high speeds irrespective of lighting and weather. Therefore, the system was evaluated under real velodrome conditions using quantitative measures including lap-detection accuracy and timing precision as well as qualitative feedback from track cyclists on setup, usability and performance. This project demonstrates how mobile devices and computer vision can be used to provide accessible ubiquitous solutions for athletics.

ACM Reference Format:

Dasarath Katragadda, Elias Lind, Miguel Schlicht, and Haidyn Arnett. 2026. Low-Cost Velodrome Lap Timing via iPhone and Computer Vision: Report. 1, 1 (May 2026), 6 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Track cycling at velodromes requires accurate lap timing for training and performance evaluation. However, existing solutions are inadequate for individual training use. Race-grade timing systems are prohibitively expensive, manual stopwatch methods are imprecise and require additional personnel, and GPS-based approaches are unreliable in indoor or multipath-heavy environments. As a result, there is currently no affordable and accessible lap-timing solution for track cyclists training independently.

To address this gap, we proposed a smartphone-based computer vision system that uses a handlebar-mounted iPhone to detect finish line crossings and compute lap splits. The system leverages a

custom-trained YOLOv26n model for real-time detection and provides immediate audio feedback to the rider, allowing eyes-free interaction during high-speed intervals (20–40 mph). This work explores principles of ubiquitous computing for athletics, emphasizing low-cost sensing and minimal-attention interaction design. Compared to our initial proposal of a fixed, finish-line mounted camera, we pivoted to a rider-mounted approach to improve portability and usability. Overall, this project demonstrates how consumer devices can democratize performance tracking technology for amateur athletes.

2 Background

Track cycling takes place on banked velodromes using fixed-gear bicycles, where riders complete structured training efforts that rely heavily on lap timing for performance evaluation. Accurate lap splits are essential for pacing, effort control, and measuring improvement across sessions.

Despite the high level of instrumentation in modern cycling, lap timing on the track remains largely unsupported. Cyclists commonly use power meters, heart rate monitors, and GPS for road cycling, but these tools do not translate well to the velodrome environment, where laps are short and conditions differ significantly from open-road settings [Passfield et al. 2016].

Existing timing solutions are not well-suited for individual training. Race-grade systems combine magnetic loops, transponders, and high-speed cameras to achieve high accuracy, but they are expensive and inaccessible for everyday use [Lynx System Developers, Inc. 2026; MYLAPS 2026]. Manual timing with a stopwatch is imprecise, requires additional personnel, and does not provide reliable real-time feedback to the rider. GPS-based approaches are also ineffective due to indoor environments and multipath effects, which introduce positional errors too large for accurate lap timing [Alahdal et al. 2020; Subirana et al. 2011].

Prior work in sensing highlights both the potential and limitations of alternative approaches. Indoor localization methods such as ultra-wideband and RFID-based systems require significant infrastructure and can suffer from missed detections or limited feedback capabilities [Kolaja and Ehlerova 2019; Minne et al. 2019]. Computer vision has been widely applied in sports for object detection and tracking, often using models such as YOLO, but these systems typically rely on stationary cameras rather than mobile, rider-mounted perspectives [Thomas et al. 2017].

From an interaction design standpoint, high-speed athletic contexts impose strict constraints on user attention. Research shows that visual distraction can negatively impact performance and safety, particularly in tasks requiring strong peripheral awareness [Wood

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ACM XXXX-XXXX/2026/5-ART

<https://doi.org/XXXXXXX.XXXXXXX>

et al. 2006]. As a result, effective systems in this domain prioritize eyes-free interaction, simple event-based input, and feedback mechanisms such as audio that minimize cognitive load.

Our work builds on these insights by introducing a low-cost, rider-portable lap timing system using a handlebar-mounted smartphone. The system leverages a custom-trained YOLOv26n model to detect finish line crossings from a moving perspective and provides real-time audio feedback to the rider. By combining accessible hardware, mobile computer vision, and low-attention interaction design (e.g., audio-based lap feedback), the system integrates with existing training workflows (e.g., Strava) while remaining practical for everyday use.

3 Implementation

3.1 System Architecture

With accessibility and price in mind, the system design chosen only requires an iPhone and a handlebar phone mount. The phone is mounted with the camera facing downward and slightly tilted up so that when recording the front tire is visible and there is some space at the top of the screen. On the software side, there are three main components to the system: the mobile app, the computer vision pipeline, and the Strava integration. The iOS mobile app utilizes the device's camera to record a live feed of the surface of the velodrome. The video is fed frame by frame into the detection model and determines if the start/finish line (see is present. If the line is present, the app laps and records the time for that lap. Additionally, the speaker on the phone announces both the lap number and time to the rider. All activities are stored on device to protect privacy of users, but the data can be exported to a user's Strava account.

3.2 Computer Vision Pipeline



Fig. 1. The finish line at Dick Lane Velodrome

We selected YOLOv26n, a lightweight (nano) variant of the YOLO family, to enable real-time performance on mobile hardware. The model was custom-trained on a dataset collected from video footage at the Dick Lane Velodrome, where frames were extracted from a handlebar-mounted perspective. These frames were annotated to label finish line regions across a range of lighting conditions, camera angles, and riding speeds. Training included standard practices such

as dataset splitting, augmentation, and hyperparameter tuning, with final performance evaluated using metrics such as mean average precision (mAP) and inference time on iPhone hardware.

The detection pipeline operates on a live video stream captured by the device camera at a consistent frame rate (60 FPS). Each frame is preprocessed through resizing and normalization before being passed into the YOLO model for inference. Post-processing steps include applying a confidence threshold and non-max suppression to filter detections. Lap timing is triggered when the finish line is detected crossing a defined region in the frame, with additional debouncing logic implemented to prevent double-counting from consecutive frames. This pipeline enables reliable, real-time lap detection under high-speed riding conditions.

3.3 Mobile Application

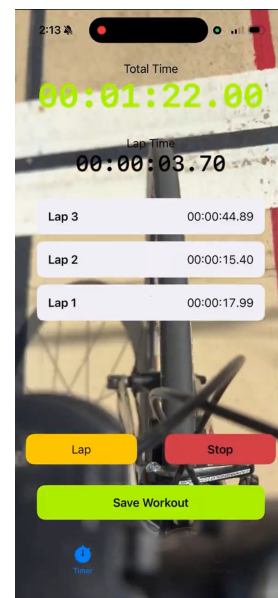


Fig. 2. Main screen with elapsed time, lap times and live camera feed

The mobile application was developed for iOS using Swift and SwiftUI. It serves as the primary interface for controlling timing efforts and processing sensor data in real time. The app captures a live video feed from the device camera and runs the YOLO-based detection pipeline frame by frame to identify finish line crossings. When a crossing is detected, the system records a lap, computes the split time, and immediately provides audio feedback using text-to-speech, allowing riders to receive updates without visual attention. The interface also displays key information such as current lap count, most recent lap time, and average pace, while all ride data is stored locally on the device for later access.

To support this functionality, the phone is mounted securely on the handlebars using a standard bike mount, with the camera oriented downward and slightly angled to capture the track surface and finish line markings. This positioning ensures consistent visibility of the finish line while maintaining stability during high-speed riding.

3.4 Strava Integration

To integrate with existing training workflows, the system includes support for Strava through OAuth-based authentication with the Strava API. After a ride is completed, the application uploads the session as a structured activity, including detailed lap split data. Each lap is recorded as an individual segment with its corresponding timing, enabling users to analyze performance directly within Strava. Additional metadata such as velodrome location, bike type, and training notes are included to provide context and ensure compatibility with standard fitness tracking practices.

3.5 Linking to Original Motivation

The system was designed to directly address the goals outlined in the original proposal, particularly achieving high-resolution lap timing, reliable detection, low cost, and safe real-time interaction. Our target of ≤ 100 ms lap-time measurement resolution was approached through high frame-rate video capture and efficient on-device inference, enabling precise timestamping of finish line crossings. To ensure reliable lap detection, the system combines a custom-trained YOLOv26n model with post-processing logic such as confidence filtering and debouncing, reducing the likelihood of missed laps or false positives during continuous motion.

A key design objective was accessibility. By leveraging an existing iPhone and requiring only a low-cost handlebar mount, the system avoids the need for expensive race-grade infrastructure while maintaining practical performance for training use. Finally, the use of real-time audio feedback allows riders to receive lap splits without diverting visual attention, supporting hands-free, eyes-free interaction during high-speed efforts. This directly addresses the safety and usability concerns identified in the original proposal, reinforcing the system's suitability for real-world training conditions.

3.6 Challenges Addressed

The system was developed to address several key challenges inherent to velodrome lap timing. One primary challenge is high-speed motion (20–40 mph), which requires fast and accurate detection under continuous movement. To address this, we use the YOLOv26n model optimized for real-time performance, combined with high frame-rate video capture and debouncing logic to ensure stable and precise lap detection.

Another challenge arises from the mismatch between start and finish locations, as riders often begin efforts at arbitrary points on the track. To handle this, the application includes manual start and stop controls, allowing riders to initiate timing independently of their physical position on the track.

Rider distraction and safety also present a significant concern, as cyclists must maintain focus and peripheral awareness at high speeds. To mitigate this, the system provides audio-only feedback during active riding, enabling eyes-free interaction, while limiting visual interaction to a glanceable display when it is safe to do so.

Environmental variability, particularly changing lighting conditions at the velodrome, introduces additional complexity for computer vision. This was partially addressed through data augmentation during training and by collecting data across multiple sessions and conditions to improve model robustness. This remains partially

addressed because we would need more varied data to properly train our model to consistently perform under different weather conditions.

Limited access to the velodrome for testing posed another constraint on development. To overcome this, much of the system was developed and validated offline using pre-recorded video, with live testing sessions concentrated into limited available time windows.

Finally, the system design evolved from the original proposal. We pivoted from a fixed, finish-line mounted camera to a handlebar-mounted approach to improve portability and eliminate the need for external infrastructure. While this introduced the added challenge of detection from a moving camera, it ultimately resulted in a more practical and user-friendly solution for real-world training.

3.7 Technical Novelty

This work introduces, to our knowledge, the first handlebar-mounted computer vision lap timing system for track cycling. Unlike prior approaches that rely on fixed infrastructure or stationary cameras, our system performs real-time detection from the rider's perspective, enabling a fully portable and self-contained solution for training. This required custom training of a YOLOv26n model specifically for finish line detection under dynamic conditions, including varying angles, motion, and lighting encountered during high-speed riding.

Beyond sensing, the system integrates computer vision-based timing directly with a consumer fitness platform (Strava), creating a seamless workflow for recording and analyzing training data. Additionally, the interaction design is optimized for high-speed athletic contexts through the use of real-time audio feedback, enabling eyes-free operation and minimizing rider distraction. Together, these contributions demonstrate a novel combination of mobile computer vision, accessible hardware, and low-attention interaction design tailored to real-world athletic use.

4 Results

4.1 Quantitative Evaluation

4.1.1 Lap Detection Reliability. The overall lap detection reliability was extremely high, with the system correctly detecting every lap without fail. Across 2 days of testing, approximately 20 full laps were completed at race pace, with each lap being detected reliably and instantaneously.

While the incidence rate of false negatives was zero, there were a two false positives when the users failed to turn off the tracking before exiting the velodrome, and where the users started the tracking before entering the velodrome. The system tended to output a false positive when provided with horizontal lines with color gradients. While this is technically an issue, the scenario was outside of the scope of the system's intended use case, and thus can be ignored for this reliability analysis.

4.1.2 System Performance. The overall system's performance was extremely high due to a combination of factors. The model was implemented using a YOLOv26n backbone. This model is the "nano" version of YOLO's object detection suite, and is the most lightweight mobile compute optimized version of YOLO's models. This model architecture provided the capability for on-device processing, which was further optimized through Apple's CoreML machine learning

framework. This framework allows for the deployment of ML Models on the on-device AI Compute Engines, which heavily accelerates performance over standard CPU compute. This resulted in the application maintaining a consistent 60fps across all testing scenarios, which is simply what the camera acquisition rate was set to. This correlates to a guaranteed 17ms resolution, as that is the temporal spacing between processed frames.

4.1.3 Timing Accuracy. Due to this resolution time, we have partially achieved our goal of a sub 100ms resolution time. The only missing component is that timing does not currently detect when the finish line crosses the front wheel, but instead when the finish line enters the frame. This discrepancy can offset timing by a varying number of milliseconds depending on speed. However, this can easily be remedied in a future version by adding wheel plane detection or manual demarcation.

4.2 Qualitative Evaluation

4.2.1 User Study Setup. In order to test the application’s effectiveness, we recruited cyclists from the Georgia Tech Cycling Team to provide feedback on the system. Each cyclist was allowed to test the application as much or as little as they desired, after which they were able to provide feedback regarding their experiences.

4.2.2 Usability Feedback.

- Setup ease: 4.75/5
- Readability during training: 3.33/5
- Cognitive load: 1.83/5
- Timing accuracy & reliability: 4.75/5
- Overall satisfaction: 4.83/5

Overall, the cyclists agreed that the application was extremely simple to use and reliable for lap time detection. They also agreed that the application was significantly more convenient than any other pre-existing timing systems. However, some of the testers mentioned that they would like to see expanded functionality in a future version. This includes syncing Power, Cadence, and Heart Rate sensor data in order to provide a complete overview of the training data. Similarly, some users mentioned that the popup for lap splits was slightly small and difficult to read, and that a full screen popup would be helpful.

4.3 Figures and Tables

Table ?? summarizes the usability feedback in more detail.

Table ? presents a summary of key system performance metrics observed during testing. As mentioned earlier, the system achieved near-perfect lap detection reliability under normal usage conditions and operated at a consistent 60 FPS, corresponding to a timing resolution of approximately 17 ms.

5 Conclusion and Reflection

5.1 Project Summary

This project developed a low-cost lap timing system for track cycling using a handlebar-mounted iPhone and a custom-trained YOLOv26n computer vision model. The system leverages the phone’s camera to detect finish line crossings in real time and compute lap splits directly on-device, eliminating the need for external infrastructure.

Metric	Mean Score	Interpretation
System performance	4.83	Very strong
Lap timing accuracy	4.50	Strong
Lap detection reliability	5.00	Excellent
Usability	4.83	Very strong
Ease of setup	4.67	Very strong
Cognitive load	1.83	Low effort
Readability during training	3.33	Moderate
Overall satisfaction	4.83	Very strong
Preference over traditional timing	4.40	Strong preference

Table 1. Summary of user feedback scores. All metrics were rated on a 1–5 scale. For cognitive load, lower scores indicate lower perceived effort.

Metric	Result
Lap detection reliability	~100% (no missed laps in testing)
False positives	2 (outside normal usage conditions)
Frame rate	60 FPS
Timing resolution	~17 ms
Real-time feedback latency	Near-instantaneous
Setup time	<2.5 minutes

Table 2. System performance summary based on testing observations.

In testing, the system achieved near-perfect lap detection reliability, correctly identifying all laps during controlled sessions, with only minor false positives occurring outside of normal usage conditions. Timing resolution approached the target sub-100 ms goal, with a frame-based resolution of approximately 17 ms, though minor offsets remained due to detecting line entry rather than exact wheel crossing.

Beyond detection performance, the system was successfully integrated with Strava, allowing users to export lap data seamlessly into their existing training workflows for analysis and sharing. A key design contribution of this work is its emphasis on eyes-free interaction, with real-time audio feedback enabling cyclists to receive lap information without diverting visual attention during high-speed efforts. Importantly, the entire system requires only a smartphone and a simple handlebar mount, making it orders of magnitude cheaper than traditional race-grade timing systems while still providing practical accuracy and usability for training contexts.

5.2 What Went Well

Several aspects of the project were particularly successful across both technical performance and user experience. The computer vision approach proved highly effective, as the lightweight YOLOv26n model was able to reliably detect finish line crossings in real time from a moving, handlebar-mounted perspective—an especially challenging setting due to speed and camera motion. The audio feedback system was also well received, with users reporting that it was clear, non-distracting, and enabled true eyes-free interaction, which is critical in a high-speed athletic environment.

The portability of the system emerged as a major strength, as the handlebar-mounted design allowed cyclists to use the system at any velodrome without requiring setup of external equipment.

User feedback further reinforced the system's practicality: cyclists rated overall satisfaction highly (4.83/5) and found the system significantly more convenient than manual stopwatch timing methods. Additionally, the integration with Strava fit naturally into existing training workflows, making recorded data immediately accessible, analyzable, and shareable.

Finally, as a team, we were able to work well together and distribute our skills to various portions of the project to effectively complete our goals.

5.3 What Did Not Go Well / Challenges Encountered

Despite strong overall performance, the project encountered several limitations and challenges throughout development and evaluation.

One of the primary constraints was limited access to the velodrome, which made it difficult to schedule testing sessions and slowed iteration cycles. As a result, much of the system had to be developed and validated using pre-recorded video, which did not fully capture the complexities of real-world riding. Environmental variability also posed challenges, particularly changes in lighting conditions such as shadows and indoor-outdoor transitions, which occasionally impacted detection robustness.

Additionally, the training dataset for the YOLOv26n model was limited to footage from a single velodrome (Dick Lane), raising concerns about generalizability to other tracks with different markings, surfaces, or lighting conditions. While the system performed reliably under typical conditions, certain edge cases—such as multiple riders in close proximity, tightly grouped packs, or extreme lean angles—introduced potential failure modes that were not fully addressed.

Finally, there was a relatively small user testing sample size due to limited access to experienced track cyclists.

Together, these challenges highlight areas for improvement in both system robustness and evaluation breadth in future iterations.

5.4 Achievement of Project Objectives

The project largely met its core objectives, with strong performance in lap detection, usability, and system integration. Reliable lap detection was successfully achieved, with the system correctly identifying all laps during controlled testing sessions and only minor false positives occurring outside of intended usage scenarios. The goal of achieving ≤ 100 ms timing resolution was partially met: while the system operated at approximately 17 ms frame resolution due to 60 FPS processing, timing accuracy was affected by detecting when the finish line entered the frame rather than when it crossed the front wheel, introducing a small but variable offset.

Real-time feedback was fully achieved, with low-latency audio output providing immediate lap information to riders. User feedback indicated that this feature was both effective and essential for maintaining focus during high-speed efforts. The low-cost objective was also achieved, as the system requires only a smartphone and a handlebar mount, making it significantly more accessible than traditional race-grade timing systems.

In terms of usability, the system performed strongly, with high user satisfaction ratings (4.83/5) and positive feedback regarding ease of setup and reliability. While some users noted minor interface

limitations (e.g., readability of on-screen text), the overall experience was considered practical for real training use. Taken together, these results indicate that the project objectives were largely met, with remaining gaps primarily related to edge-case robustness and precise timing accuracy. These outcomes reflect key design tradeoffs made during development, particularly prioritizing accessibility, real-time performance, and usability over perfect measurement precision.

5.5 Broader Implications

This work highlights the potential for democratizing performance tracking in athletics by leveraging widely available consumer devices and modern computer vision techniques. By replacing expensive race-grade infrastructure with a smartphone-based solution, the system demonstrates how advanced sensing capabilities can be made accessible to amateur athletes, lowering the barrier to entry for high-quality training tools. More broadly, the project illustrates key principles of ubiquitous computing in athletic contexts, particularly the importance of context-aware, low-attention interaction design. The use of real-time audio feedback enables effective eyes-free interaction, which is critical in high-speed, safety-sensitive environments such as track cycling.

The approach also has strong potential for generalization beyond velodrome cycling. Similar computer vision-based timing systems could be applied to other domains, such as running tracks, swimming laps, or skiing gates, where lap or segment timing is important but infrastructure is limited. In addition, this work contributes to the growing DIY sports technology movement, empowering athletes to build and customize their own training tools rather than relying solely on commercial solutions.

From an interaction design perspective, the project reinforces the importance of minimizing cognitive load and visual distraction in mobile systems used in dynamic, real-world environments. At the same time, it raises open questions about how to balance accuracy, cost, and usability in sports sensing systems. While this project prioritizes accessibility and real-time feedback, future work must consider how to further improve precision without sacrificing the simplicity and portability that make such systems valuable.

5.6 Future Work

A primary limitation of the current system lies in the generalizability of the computer vision model, which was trained exclusively on data from the Dick Lane Velodrome. As a result, performance may degrade when applied to other velodromes with different track markings, surfaces, or lighting conditions. Future work should prioritize collecting and incorporating data from multiple velodromes to improve robustness and enable reliable deployment across diverse environments.

There are also significant opportunities to expand the system's functionality and user experience. For example, predictive pacing features could leverage historical lap data to provide real-time guidance to riders, while integration with external sensors such as power meters, cadence sensors, and heart rate monitors would offer a more comprehensive view of athlete performance. Furthermore, live coaching capabilities could be enabled by transmitting lap data

in real time to coaches, allowing for immediate feedback during training sessions.

From an evaluation standpoint, the system would benefit from a larger-scale user study involving more track cyclists across multiple velodromes. This would provide stronger validation of usability, reliability, and real-world adoption potential, as well as uncover additional edge cases and improvement opportunities.

Finally, open sourcing the application presents an opportunity to accelerate development and adoption within the cycling community. By enabling athletes and developers to contribute improvements, the system could evolve into a more robust, widely used tool for track cycling training.

5.7 Lessons Learned

Several key lessons emerged throughout the development of this project. One of the most important was the value of early user testing. The decision to pivot from a fixed camera setup to a handlebar-mounted design was driven directly by user feedback, and identifying this need earlier would have saved development time. More broadly, working closely with track cyclists reinforced the importance of domain expertise, as many practical requirements (e.g., the need for eyes-free interaction and minimal cognitive load) only became apparent through real user engagement.

We also learned that real-world testing is essential, particularly for systems operating in dynamic environments. While offline development using pre-recorded video was useful for initial prototyping, it failed to capture critical challenges that only appeared during live riding, such as lighting variability, readability, and rider movement. Additionally, this project highlighted the inevitability of tradeoffs in system design. Balancing accuracy, latency, cost, and usability required deliberate decisions, especially given the focus on training rather than race-grade precision.

Finally, the project made it clear to us that interaction design is just as important as sensing performance. Achieving accurate lap detection alone was not sufficient; equal attention had to be given to how information was delivered to the user. The incorporation of audio feedback proved to be a crucial design decision, enabling low-attention interaction and making the system viable in a high-speed, safety-critical context.

5.8 Final Reflection

Overall, this project can be considered a strong success. The system not only functioned as intended, but also addressed a real and previously unmet need within the track cycling community. Notably, users expressed genuine interest in the application and began incorporating it into their training workflows, demonstrating its practical value beyond a purely academic exercise.

From a personal and team perspective, the project provided meaningful growth in understanding how real-world constraints shape technical solutions. Interacting directly with users immediately after building features offered rapid feedback loops and made the development process both more effective and more rewarding. The experience also highlighted the importance of designing for cognitive load, particularly in high-speed environments, which ultimately led to key decisions such as prioritizing audio feedback over visual

interaction. Additionally, working in the sports technology domain proved to be especially engaging, reinforcing the value of building systems that can be tested and iterated on directly in their intended context.

The project also has the potential to make a broader impact within the track cycling community, particularly as a low-cost and accessible alternative to existing timing solutions. Among the team's proudest achievements is the fact that we built a system that people actually use, applied our technical skills to solve a real problem, and made the best use of our time throughout the semester.

If given the opportunity to revisit this project, we would focus on expanding testing across multiple velodromes and increasing the size and diversity of the user base. Although we are very satisfied with the result, a greater validation would strengthen confidence in the generalizability of the system and the long-term adoption potential.

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